# Exploring Machine Learning and Nonlinear Dynamics in Large Language Models

# Introduction & Motivation

- LLMs can behave like evolving systems, producing sequences that unfold over time
  - Generate token-by-token trajectories
  - Can show drift, collapse, oscillatory cycles, and instability
  - Recent work: LLMs exhibit attractors, multistability, and regime switching
- Understanding dynamics helps explain hallucinations, loops, temperature sensitivity, and other unstable behaviors

# Main Research Questions

How does temperature reshape GPT-2's attractor landscape?

How does desirability relate to attractor stability?

How do seeds behave as initial conditions?

Do the observed patterns match predictions from nonlinear dynamical systems theory?

# Background

## 1. ML & Dynamical Systems

ML models can learn or interact with nonlinear dynamical systems [1], [2]

## 2. LLMs & Dynamical Reasoning

LLMs can infer rules of dynamical systems from sequences [3]

## 3. Attractor Behavior in LLMs

Repeated generation can converge to fixed points or cycles [4]

## 4. Sensitivity to Initial Conditions

LLM outputs vary under different seeds/prompts [6], [7]

## 5. Symbolic Dynamics as a Tool

State discretization & transition analysis are standard in dynamical-systems research [5]

# Pipeline Overview

Mini-Lab Code → NLP Portion → ML Portion → Symbolic Dynamics → Aggregation

# Setup: Data Collection

- Raw GPT-2 data generated using Mini-Lab code [8]
- Purpose: create a controlled dataset to feed into analysis pipeline
- Parameters:
  - Prompts: Jesus & Moon Landing (real, fake, real or fake, fake or real)
  - Temperatures: 0.001, 0.300, 0.500, 0.700
  - Seeds: 1, 2, 3
- Output: 15 txt files

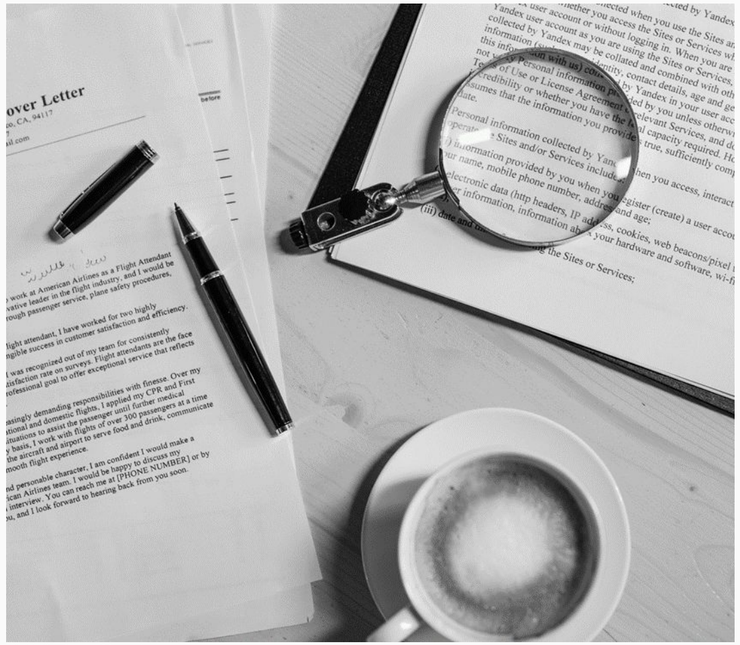
```
# === EDIT HERE ===
PROMPT = "Is Jesus Christ fake"
MODEL = "gpt2"            # e.g., "gpt2", "distilgpt2"
TEMPERATURE = 0.7
TOP_K = 50
MAX_NEW_TOKENS = 300
SEED = 1

# Similarity (sentence -> symbol)
NGRAM_MIN = 3
NGRAM_MAX = 5
SIM_THRESHOLD = 0.45

# Optional: input your own text instead of generating
INPUT_TEXT = ""  # paste your own text here (leave empty to generate with GPT-2)
```

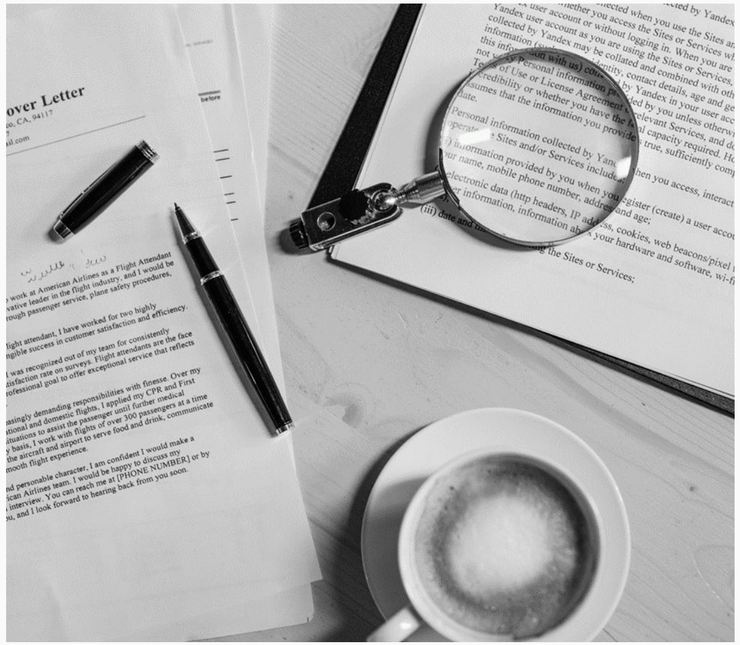
# Setup: NLP

- Main goals:
  - Sentence segmentation (assumes one sentence / response per line)
  - TF-IDF char-level vectorization
  - KMeans clustering → symbolic states
  - coherence / stability / perplexity metrics
  - Saves embeddings, states, quality scores



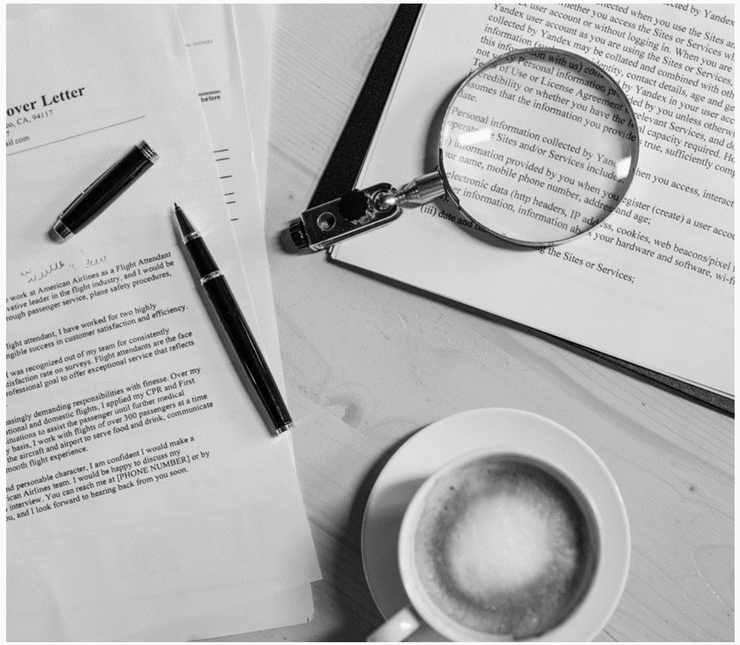
# Setup: Machine Learning

- Main goals:
  - Take NLP outputs (perplexity, length, stability, symbol)
  - Define/classify "desirable" = low perplexity + low instability
  - Train logistic regression to predict desirability
  - Save per-sample predictions in ml_output.csv, with Q_class
- Model performance:
  - Mean accuracy: 0.643
  - Std. deviation: 0.212
  - Accuracy vs. temperature:
    - High accuracies at extremes (0.72 at t = 0.001, 0.7) lowest at mid range (0.48 at t = 0.3 & 0.5 at t = 0.7)



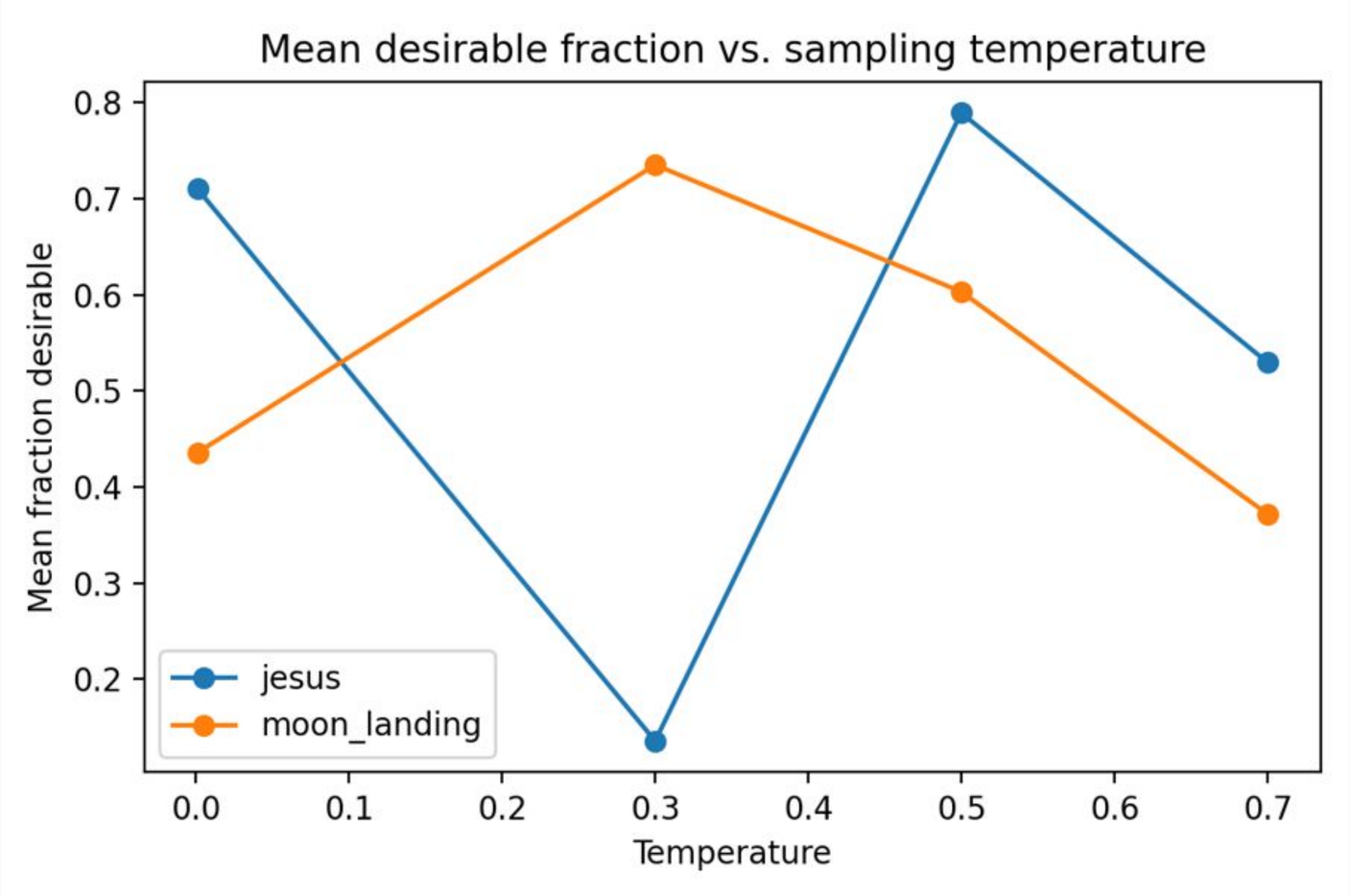
# Setup: Dynamics

- Main goals:
  - Build Markov-style transition matrix over symbolic states
  - Compute stationary distribution / spectral gap
  - Simple "dynamics type" classification (fixed, oscillatory, mixed)
  - Time-to-settle (if there is a final attractor)
  - PCA phase portrait of embeddings with flow arrows



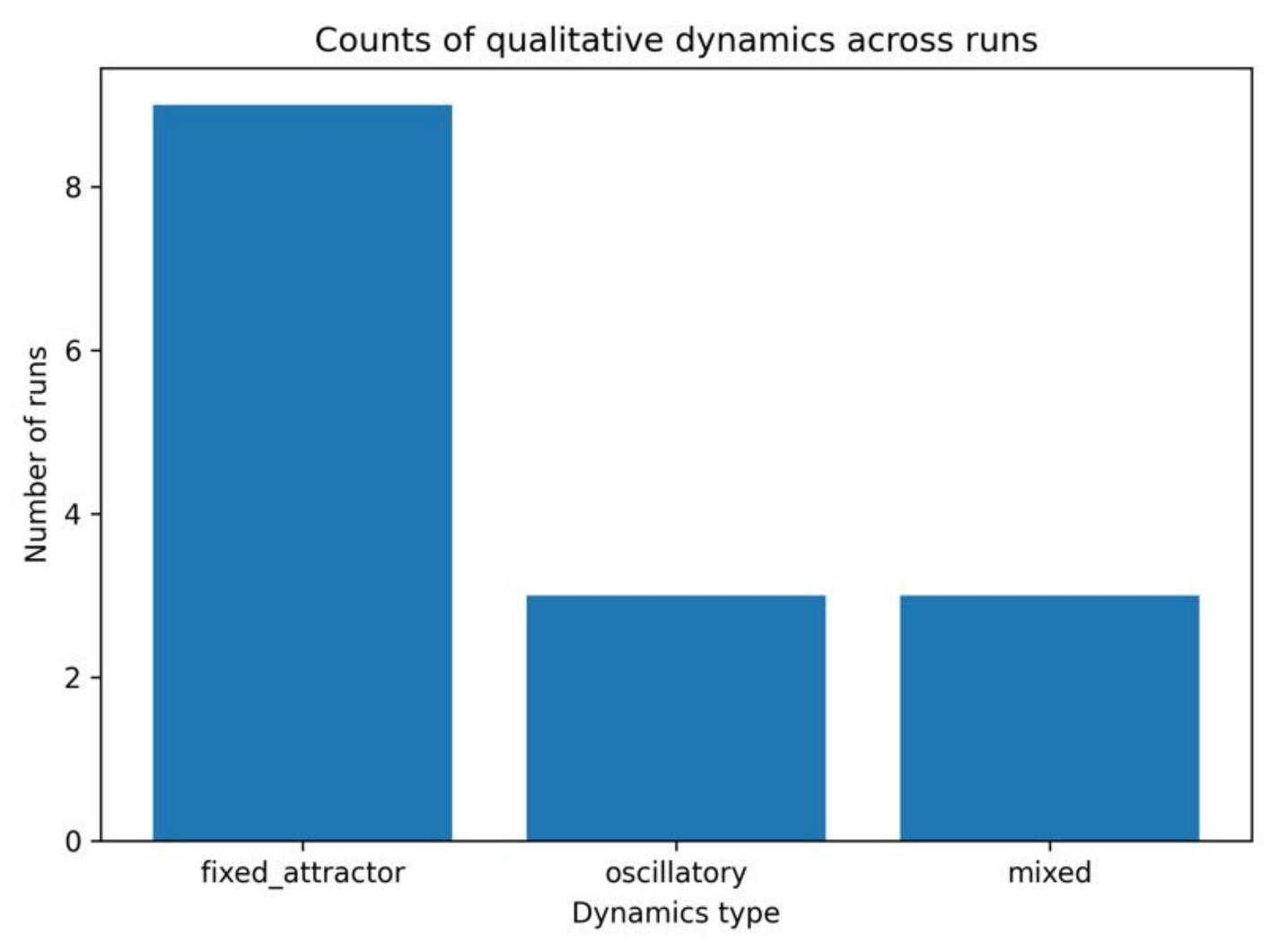
# Results: Temperature

- Jesus prompt:  high performance at T = 0.5, low at T = 0.3
- Moon landing prompt: high at T = 0.3, declines after
- Doesn’t produce monotonic changes
- Nonlinear patterns resemble bifurcations



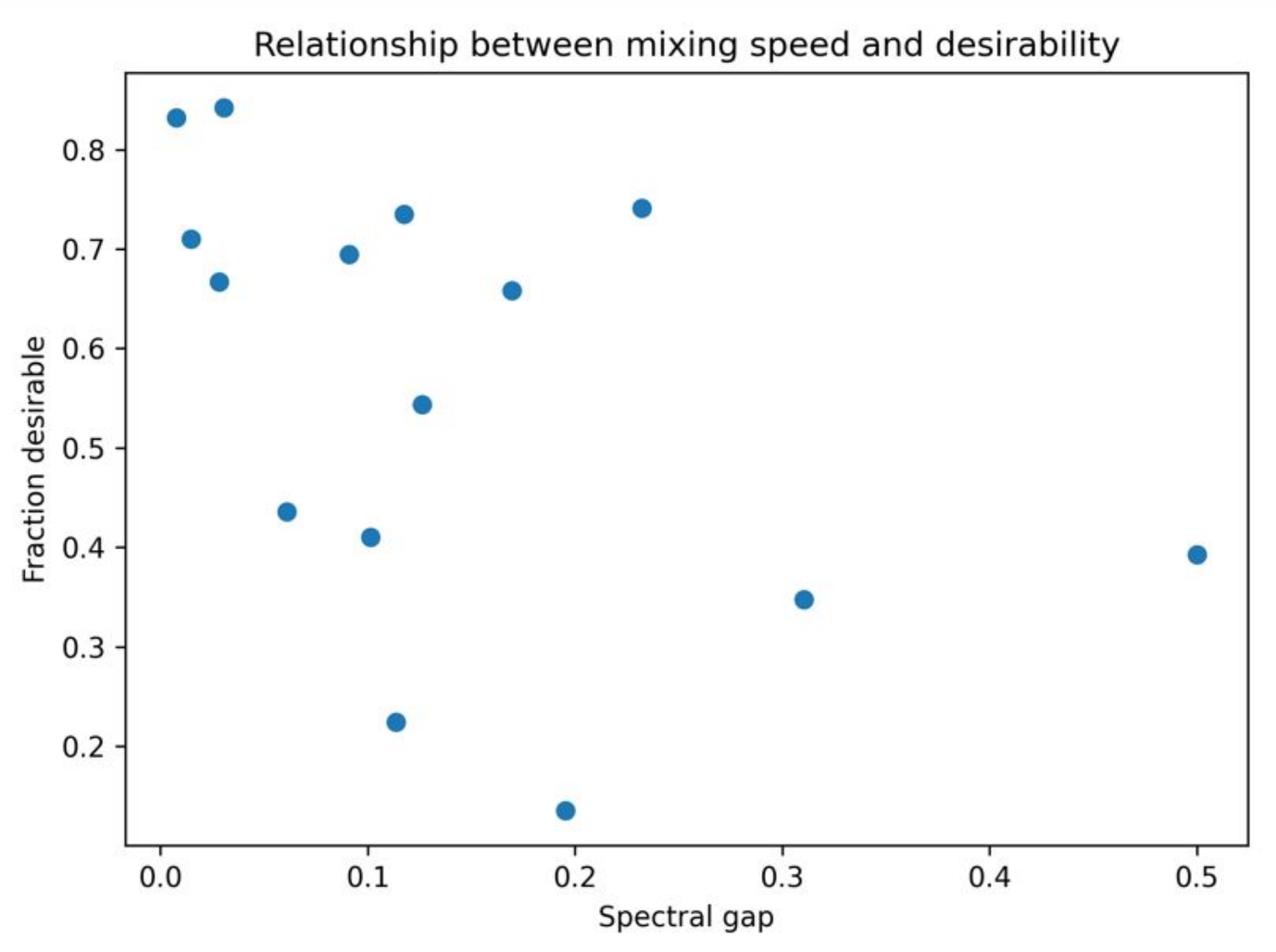
# Results: Attractor Type Distribution

- Majority: fixed attractors (9)
  - converges to a stable point and stays there
- Some oscillatory (3)
  - repeats a stable loop or cycle indefinitely
- Some mixed (3)
  - sometimes settles to a point and sometimes enters a cycle, depends on initial conditions
- Suggests GPT-2 often collapses to a simple attractor



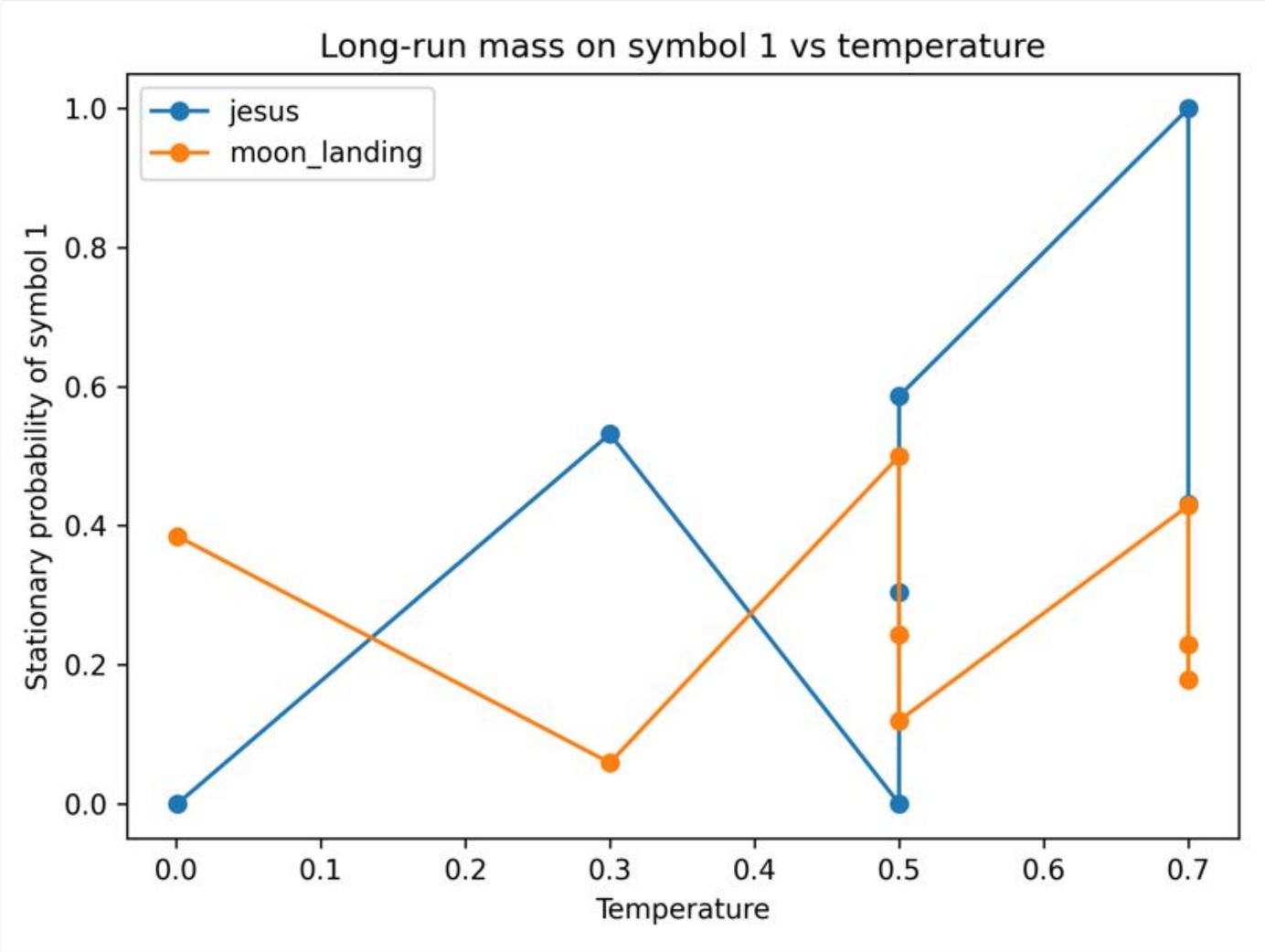
# Results: Spectral Gap vs Desirability

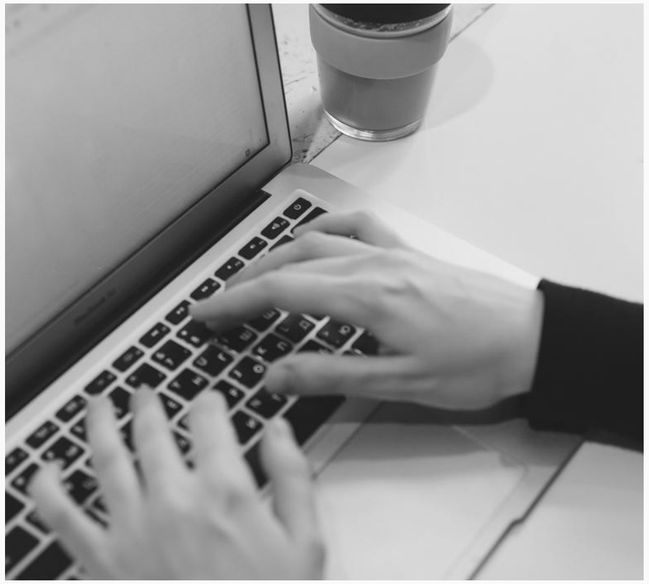
- Smaller gap = slower mixing = more desirable
- Fast-mixing dynamics → unstable or chaotic
- Desirable runs reside in stable attractor regions



# Results: Stationary Distribution Patterns

- High stationary mass → stable desirable behavior
- Temperature shifts long-run probability structure
- Some high-T conditions collapse into a single symbol





# Results: Main Points

- Prompt: initial-condition shift in starting embedding
- Temperature: control parameter,
- Seed: initial condition, sets random sampling sequence
- GPT-2 exhibits:
  - Fixed points
  - Oscillations
  - Multistability
- Regime switching
- Behavior aligns with nonlinear systems literature [1], [2], [5]
- Explicit attractor cycles match previous research [4]

# Conclusion

- GPT-2 displays nonlinear dynamical behavior
- Temperature and seed jointly determine attractor basin
- Desirability depends on attractor stability
- Symbolic dynamics is a powerful lens for studying LLMs
- Future work: entropy rate,attractor basin mapping, more models and prompts

# References

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[8] N. Johnson, “GPT-2 Mini Lab Code,” unpublished instructional material, Biophysics: Macroscopic Physics in the Life Sciences (PHYS 3127), George Washington University, 2025.

**THANK YOU!**